Supporting information of

Water adsorption on MO_2 (M = Ti, Ru and Ir) surfaces. Importance of octahedral distortion and cooperative effects.

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Table S1: Adsorption energies for the single water adsorption on the TiO_2 (110) and (011) rutile surfaces. The relative energies (ΔE) are calculated with respect to the water adsorbed in its molecular form (mol). The adsorption energies are calculated as ΔEAds = Ecomplex – Esurf – E_{H2O}, with each system optimized separately.

		PBE-D2		PBE-D3		PBE0-D2	
(h k l)	Adsorption mode	ΔEads (kJ/mol)	ΔE (kJ/mol)	ΔEads (kJ/mol)	ΔE (kJ/mol)	ΔEads (kJ/mol)	ΔE (kJ/mol)
(110)	mol	-86.9	0.0	-85.4	0.0	-96.7	0.0
(110)	diss	-52.2	34.7	-47.7	37.7	-65.5	31.2
(011)	mol	-136.2	0.0	-129.3	0.0	-134.2	0.0
(011)	diss	-140.0	-3.7	-134.5	-5.2	-139.4	-5.2

Table S2: PBE-D2 adsorption energies for the single water adsorption on different TiO_2 (110) rutile surface models. The relative energies (ΔE) are calculated with respect to the water adsorbed in its molecular form (*mol*). The adsorption energies are calculated as $\Delta E_{Ads} = E_{complex} - E_{surf} - E_{H2O}$, with each system optimized separately.

Number of layers	Adsorption mode	ΔE (kJ mol^{-1})	ΔE_{ads} (kJ mol^{-1})
4 layer	mol	0.0	-86.9
	diss	34.7	-52.2
5 layer	mol	0.0	-105.6
	diss	-8.6	-114.3
6 layer	mol	0.0	-89.8
	diss	26.1	-63.6
7 layer	mol	0.0	-95.0
	diss	4.3	-90.6

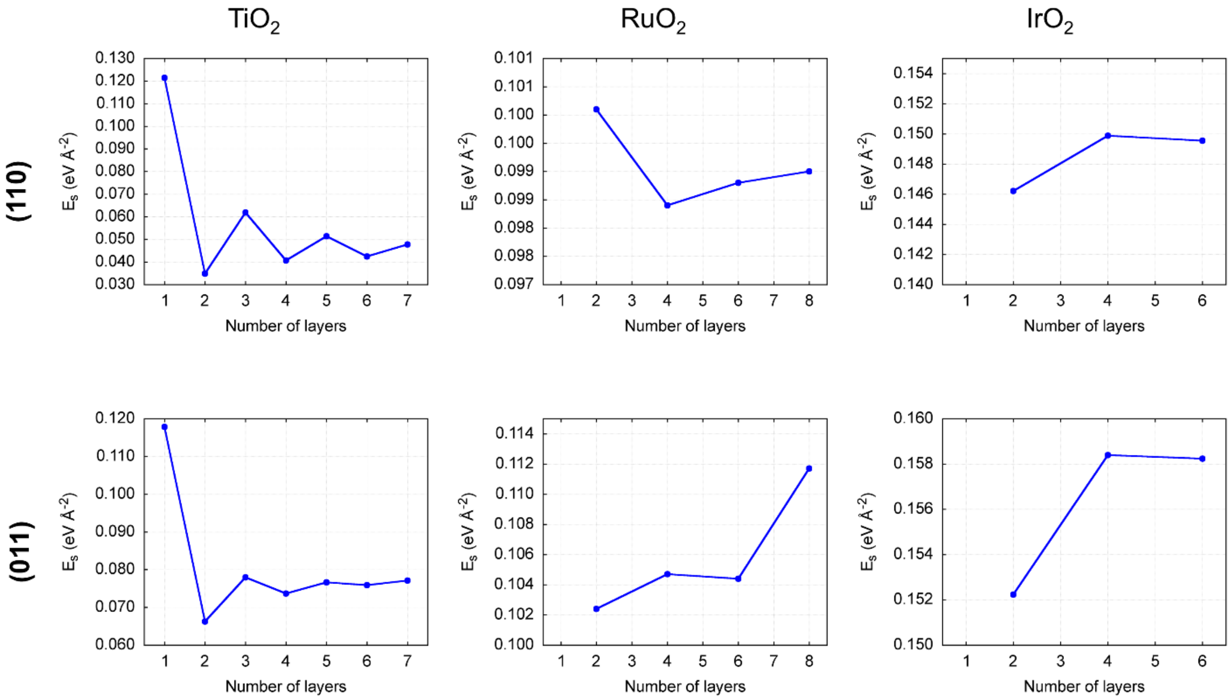

Figure S1: PBE-D2 calculated surface energies.

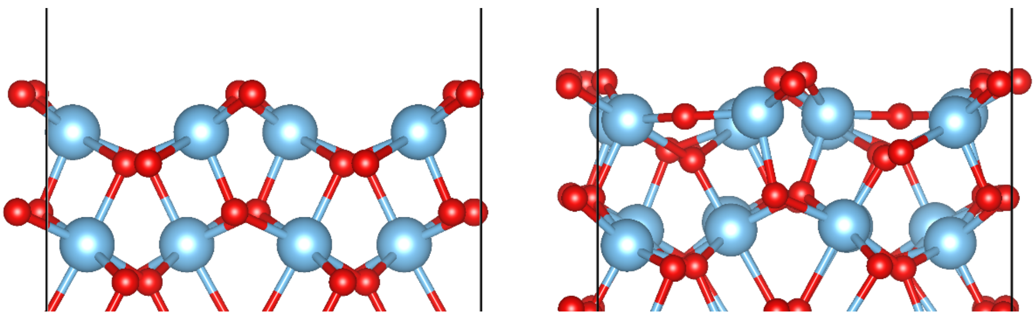

Figure S2: PBE-D2 geometries for the reconstructed (right) and non-reconstructed (left) TiO_2 (011) rutile surface.

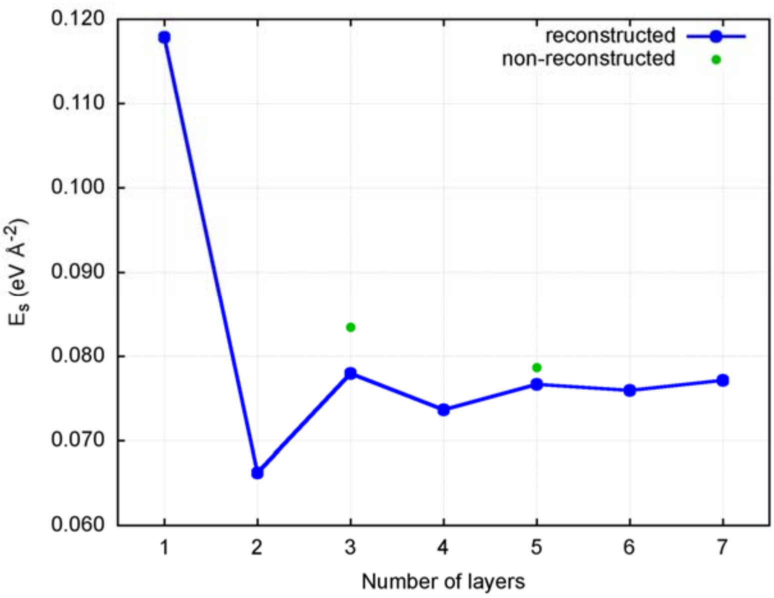

Figure S3: Surface energy for the reconstructed (blue line) and non-reconstructed (green points) TiO_2 (011) rutile surface.

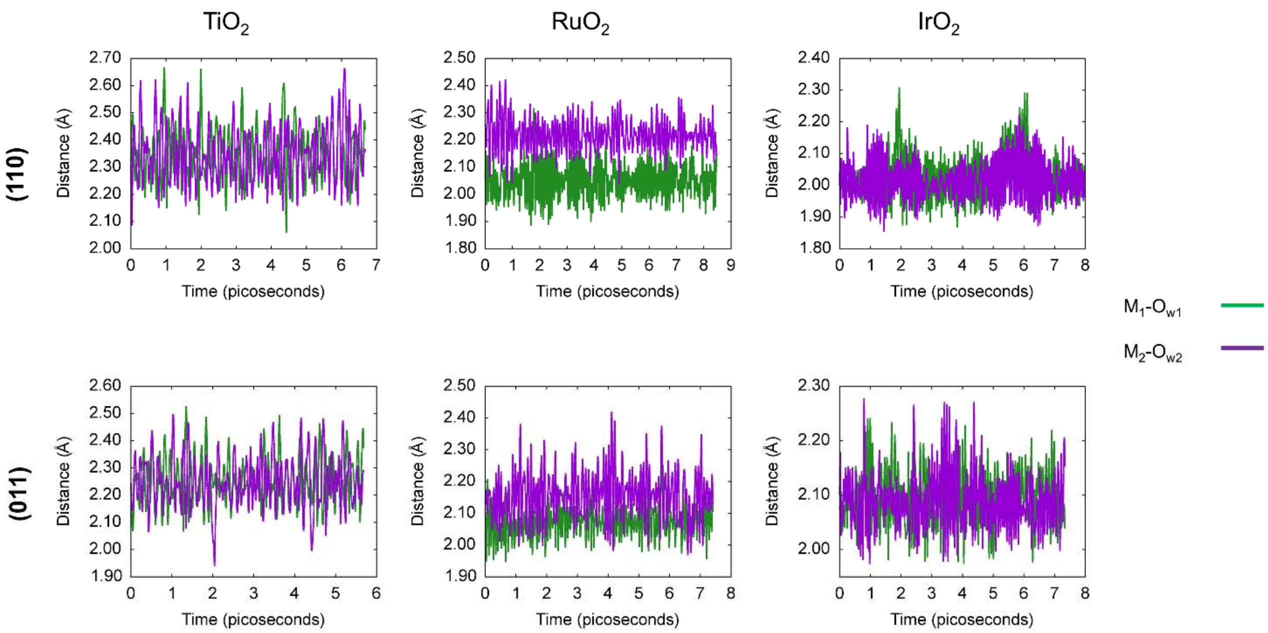

Figure S4: M—O_w distances of two neighbor water adsorbed (in Å) along the dynamics